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Effect of aerobic pretreatment on anaerobic digestion of olive mill wastewater (OMWW): An ecoefficient treatment

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ABSTRACT

Anaerobic digestion experiments of olive mill wastewater (OMWW) without pretreatment were carried out and signs of inhibition in the biological process were observed after a time equal to 1.5 times the Hydraulic Residence Time (HRT) due to the presence of a high inhibitors concentration in OMWW. Based on these findings, OMWW was subjected to an aerobic pretreatment in order to reduce the concentration of phenolic compounds and decrease Total Chemical Oxygen Demand (TCOD), achieving a reduction of 78% and 90% of the initial polyphenols concentration and 18% and 21% TCOD reduction when the substrate was aerated for 5 and 7 days respectively.

Finally, anaerobic digestion experiments using OMWW aerated for 5 and 7 days as substrate were conducted to determine the influence of aeration time on methane yields and TCOD reduction. The results yield 5 days as the aeration time that achieves best results, given that this stabilizes the anaerobic process, can reduce the TCOD by 65% and generates almost 0.39 m^3 methane/kg removed TCOD.

Anaerobic digestion of OMWW will be economically feasible if the waste is pretreated by aerobic digestion, since the period of return of investment obtained under these conditions does not exceed 6 years.

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Keywords: Aerobic pretreatment; Anaerobic digestion; Olive mill wastewater; Economic viability; Polyphenols inhibition; Methane yield

1. Introduction

Spain is the largest producer of olive oil in the world. The olive oil produced in this country is extracted by the two-phase centrifugation process, which leads to the generation of solid and liquid wastes known as Two Phase Olive Mill Wastewater (TPOMW) (Barranco et al., 2004). Such residue is subjected to several treatments: firstly, bones are separated, and subsequently pomace, olive pomace oil and olive mill wastewater (OMWW) are obtained in an olive oil mill which operates in three phases. The first two substances are value-added products suitable for commercial trading, although the latter product, a highly stable emulsion composed of vegetation water, oil and soft tissues from olive pulp, continues to cause large-scale environmental problems. Nowadays, the most common method for eliminating OMWW is through evaporation in storage ponds, owing to the low investment required and the climate conditions in Mediterranean countries (Borja et al., 2006). However, this operating method has several relevant disadvantages, such as bad odour, infiltration and insect proliferation (Roig et al., 2006). The process also produces a sludge that can be used in agriculture, but its application must be controlled in order to mitigate the increase in soil salinity (López-Piñeiro et al., 2011) and its phytotoxic effects (Magdich et al., 2012). Given that the current treatment methods do not completely solve the associated environmental problems, it is crucial to investigate new waste treatment techniques that may be both effective and economically viable.

An alternative eco-efficient treatment of OMWW, based on an anaerobic treatment preceded by an aerobic pretreatment, is reported in the present manuscript.

The main purposes of anaerobic digestion (AD) of wet waste biomass are the production of energy, the recovery of the water contained in the waste and its potential use for crop

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fertirrigation and the reuse of the solid digested effluent as an organic amendment. However, according to Hamdi (1996), AD of OMWW is limited by the inhibition of methanogenic bacteria by simple phenolic compounds (Hernández and Edyvean, 2011), certain organic acids, long-chain fatty acids (LCFA), (Gonçalves et al., 2012) and polyphenols (Sayadi et al., 2000). The inhibition associated with polyphenols has been extensively studied; however, inhibitory concentrations of these compounds are difficult to be determined given the significant differences with respect to data reported in the literature due to the dispersion of anaerobic communities (different inoculums used). However, it is clear that the effect of the toxicity is related to the molecular structure of polyphenol: the inhibitory effect is found to increase as the number of hydroxyl group decreases (Akassou et al., 2010). For such reason, a pretreatment operation is required, using a physicochemical or biological stage to decrease the organic load and the concentration of potential inhibitors of AD. Among the existing physicochemical treatments, the most prominent are the following: application of electrochemical oxidation (Belaid et al., 2013), catalytic oxidation using Fenton's reagent (El-Gohary et al., 2009) and ozonation (Chedeville et al., 2009). Alternatively, most biological pretreatments use fungus species such as *Pleurotus ostreatus* (Aggelis et al., 2003; Fountoulakis et al., 2002) and *Panus trigrinus* (DÁnnibale et al., 2004) or one of several species of yeasts, such as *Candida oleophila* (Amaral et al., 2012) and *Geotrichum candidum* (Asses et al., 2009). Moreover, several authors have applied this biological pretreatment along with an aeration in order to encourage microorganisms growth, some examples could be the use of white-rot fungus *P. triginus* (DAnnibale et al., 2006) and *P. ostreatus* (Olivier et al., 2012), or another species such as *Aspergillus niger* (Hamdi et al., 1992).

The present work reports on the use of aerobic digestion as a pretreatment operation, without adding any specific species but using the microorganisms that are naturally present in OMWW, since these are acclimated to the inhibitors concentrations that exist in the waste. This technique has been applied by others authors achieving high rates of polyphenols and organic matter degradation (El Hajjouji et al., 2008; Michailides et al., 2011).

The selection of such technology is justified by the fact that it has been proved to be cheap, effective, with low associated maintenance cost and reduced construction costs given that the evaporation ponds used in the olive oil mill industry to store the OMWW can be reused.

Before the AD experiments with OMWW were carried out, previous aerobic tests were performed in order to determine the aeration time that achieves maximum reduction in polyphenols concentration of the wastes. Next, a set of AD experiments were conducted – using OMWW as substrate with and without an aerobic pretreatment stage – in order to analyze the influence of aeration time in the anaerobic process as well as to look for the aeration period which best works from both the environmental and energy perspectives. The economic viability of the construction of an industrial-scaled AD plant to treat aerated OWM is finally assessed.

2. Materials and methods

2.1. Experimental setup for aerobic pretreatment

Fig. 1 shows the main components of the aerobic reactor. It comprises a glass flask of 2 L operating volume, the reactor

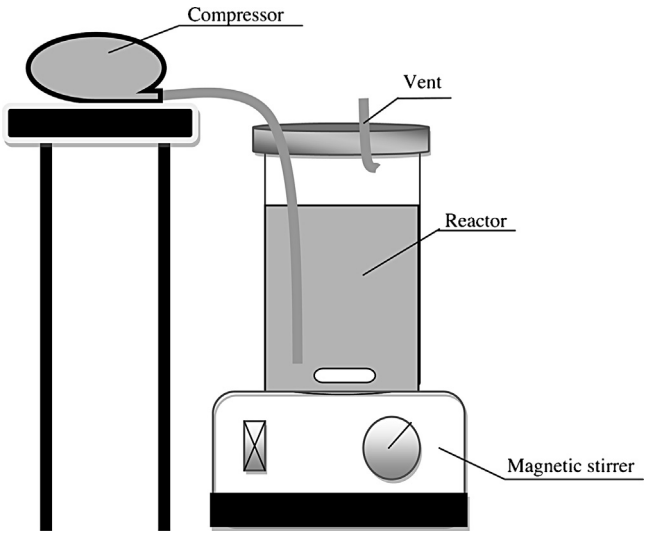


Fig. 1 – Schematic diagram of an experimental setup used for aerobic pretreatment experiments.

contents of which were homogenized using a magnetic stirrer. Air from a compressor was injected from the bottom of the reactor at a flow rate of 0.65 L air per litre of reactor and per minute. The temperature of the fermentation was not regulated.

The aeration process was conducted in discontinuous mode (in batch) without adding any specific aerobic bacterium, given that air injection activates the growth of aerobic microorganisms already present in the OMWW samples.

2.2. Experimental setup for anaerobic digestion

The anaerobic digestion experiments were carried out in a digester operating in semicontinuous mode; that is, a given volume of digested sludge was extracted daily from the reactor with a syringe and immediately the same volume of substrate was introduced, so that digestion at constant-volume configuration was guaranteed.

A scheme of the experimental setup is shown in Fig. 2, which essentially consists of a 2 L capacity glass flask, whose rim is attached to a central tube immersed in the reaction medium, and which has an input hole to introduce the substrate and an output hole to collect the biogas generated in the process. The digestion unit was submerged in a water tank and maintained at 38 °C by a thermostat equipped with a heating resistor. The substrate inside the reactor was homogenized by the continuous action of a magnetic stirrer, allowing for complete mixing.

A 5 L tank assembled to the biodigester was used to determine the volume of methane generated during the AD process. A squeeze bottle containing a sodium hydroxide solution (20% by weight) was placed between the digester and the gas tank, aimed at retaining the carbon dioxide generated during the digestion process. The methane volumes generated during the experiment displaced water in the tank, which was collected in a measuring cylinder. This way, the volume of displaced water allowed for the determination of the volume of methane generated in each experiment.

2.3. Preparation of substrate and start up of the anaerobic digestion process

OMWW, obtained through the treatment of TPOMW in an olive oil mill which operates in three phases, was periodically

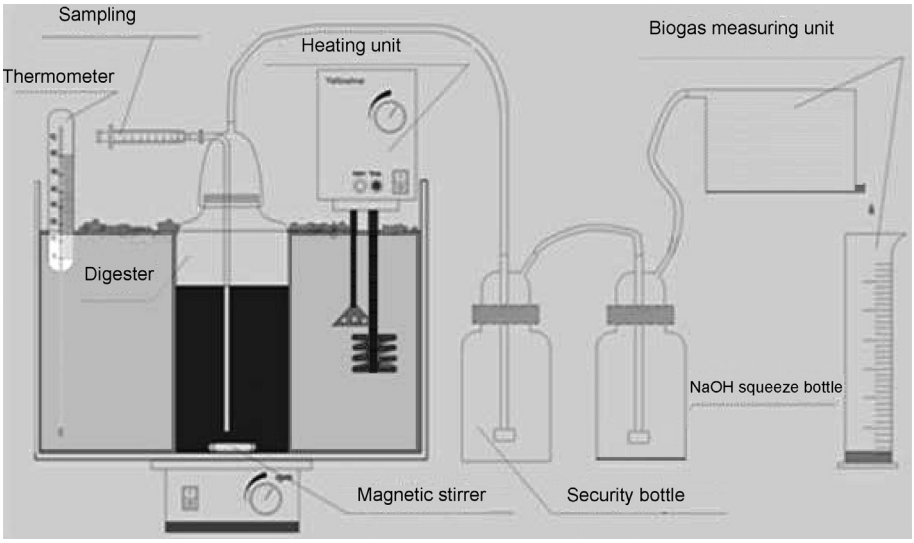


Fig. 2 – Schematic diagram of an experimental setup used for anaerobic digestion.

collected in 25 L jerry cans from TROIL Vegas Altas S.C. and the samples of aerobically pretreated OMWW were obtained in separated experiments, i.e., one aerobic essay for 5 days and another one for 7 days. All samples (both pretreated and non-pretreated OMWW) were stored at −4 °C to prevent natural degradation.

Because OMWW lacks suitable microorganisms to activate an anaerobic process, the addition of an acclimated inoculum was necessary. The sludge used as inoculum to activate biodigestion was collected from a perfectly mixed continuous anaerobic reactor located at the Wastewater Treatment Plant (WWTP) in Badajoz (Spain). This plant implemented a tertiary treatment, which was performed by anaerobic digestion, to stabilize the sludge from the primary and secondary treatments (sedimentation and aerobic treatment). After collection at the WWTP, the inoculum was transported to the laboratory and introduced into the anaerobic reactor as soon as possible. The reactor was completely filled with inoculum and sealed so that the anaerobic digestion process was guaranteed to take place in the absence of oxygen. Lastly, the process to feed the flow of substrate to be treated the next day was begun.

Each experiment was performed for a period of time that ensured that reliable results would be obtained, both in terms of the level of degradation and of methane production. The Hydraulic Residence Time (HRT) is defined as the period of time that the waste must be in contact with the bacterial population to reach its maximum degradation. This parameter was calculated as the ratio between the volume of the anaerobic reactor and the volume of substrate added daily. Therefore, a flow rate of 50 mL OMWW/day achieves a HRT of 40 days in a reactor of 2 L. The first HRT is the acclimatization period, in which the inoculum is being replaced by the OMWW, because the volume of OMWW fed is equal to the volume of inoculum extracted. In the second HRT the reaction stabilizes, the pseudo steady state of the anaerobic process was reached when methane production was stabilized and effluent TCOD was reduced and showed a constant value.

2.4. *Analytical methods*

To quantify the effects of aerobic pretreatment, a range of measurements, including pH, polyphenols concentration, Total Chemical Oxygen Demand (TCOD) and Total Solids (TS), were determined in samples taken daily during the aeration process.

Samples were taken twice weekly from the anaerobic digester in order to monitor the following parameters: TCOD, Volatile Fatty Acids (VFA), alkalinity, pH, Volatile Solids in Suspension (VSS) and Volatile Dissolved Solids (VDS).pH, VFA, alkalinity, TS, VSS and VDS were determined according to the standard methods (APHA et al., 1992). On another note, polyphenols concentration in OMWW was determined using the method proposed by Beltrán et al. (2004), whereas TCOD was analyzed using Nanocolor® (Macherey-Nagel) kits and a PF-12 portable spectrophotometer (Macherey-Nagel).

3. Results and discussion

3.1. *Aerobic digestion experiment*

The main results obtained in the aerobic digestion experiment and the characterization of the OMWW fed to the aerobic reactor, zero aeration time, are shown in Fig. 3. In Fig. 3b an increasing trend in the percentage of polyphenols degradation can be observed as aeration time was increased. The initial

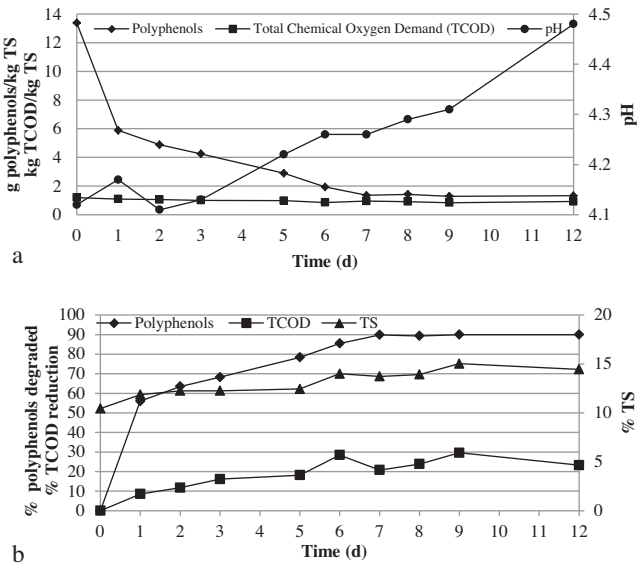


Fig. 3 – Results of aerobic digestion experiments.

polyphenols concentration was reduced by 56% after the first day of aeration, producing the greatest decrease observed. The upward trend continued over the following days of aeration, at a rate ranging between 7% and 10% decrease per day, culminating in a 90% reduction of the initial concentration on the seventh day of aeration, the highest reduction in polyphenols concentration observed. From the seventh day of aeration, the process was seen to stabilize; it therefore appears that an aeration period of longer than seven days does not lead to significant changes in the phenolic content of OMWW.

The aeration process was also seen to alter pH, TS and TCOD of OMWW. A slight increase in the pH of the OMWW was observed when the aeration time was increased, from 4.1 to 4.5 (Fig. 3a). Also the aeration process leads to an increase in the concentration of total solids, reaching stability after the sixth day (Fig. 3a). Finally, the percentage of TCOD degradation showed the same behaviour as the percentage of polyphenols degradation, i.e., an increase in the percentage of degraded TCOD was observed as the aeration time increased, to reach stability after seven days of aeration (Fig. 3b).

It should also be noted that both the polyphenols concentration and the TCOD were measured as a function of total solids concentration (Fig. 3a). The reason for this lies in the fact that if TCOD concentrations were determined as g O_2/L OMWW, a clear trend would not have been observed. Indeed it can even be expected that there will be an increase in the TCOD value over time, due to the evaporation of water in the aeration process.

Essays made by El Hajjouji et al. (2008) revealed that the treatment of the olive mill wastewater using an aerobic bioprocess could enable the efficient degradation of polyphenols. The highest polyphenols reduction, 94%, was reached with an olive mill wastewater with a C/N of 59 and a pH of 5.99. This result is similar to that achieved in this paper by treating OMWW aerated for 7 days.

Moreover, Michailides et al. (2011) developed a full-scale system for aerobic biological treatment of OMW. Continuous recirculation of the wastewater was used to provide oxygen concentrations from 0.7 to 1.2 mg/L. This system reduced 50% of phenolic concentrations under a HRT of 24 h, a higher aeration time does not allow to increase the reduction of polyphenols.

3.2. *Anaerobic digestion experiments*

Given the results of the aerobic tests, three experiments of anaerobic digestion of OMWW were conducted using as substrates:

- OMWW without pretreatment, functioning as the control sample.
- OMWW aerated for 5 days. This waste was treated for this period because the polyphenols concentration had been reduced by 78% compared to the initial sample and also because it is the HRT used in other aerobic treatments with fungi (Dhouib et al., 2005).
- OMWW aerated for 7 days. The waste was treated for this period because after this aeration time it had reached the lowest polyphenols concentration in OMWW.

It should be noted that all substrates were treated by anaerobic digestion without modifying their natural pH value.

The experiments were conducted considering 40 days HRT for all substrates; however, the organic loading rate was different in each case: 6.27 g TCOD/day for OMWW without pretreatment, 6.11 g TCOD/day for OMWW aerated for 5 days and 6.53 g TCOD/day for the OMWW aerated for 7 days.

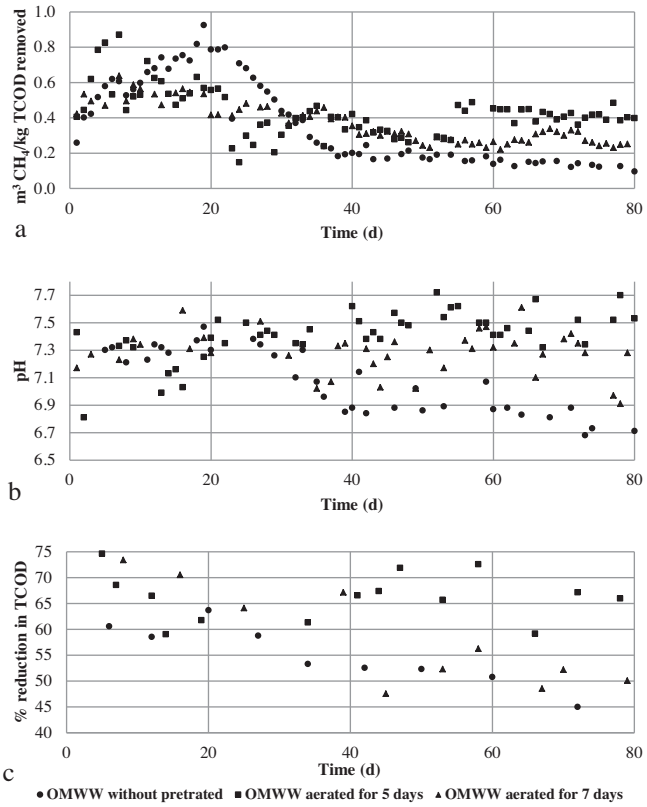


Fig. 4 – Results of anaerobic digestion of OMWW: no pretreatment stage and 5 and 7 days aeration process. (a) Methane yield, (b) pH, (c) reduction of Total Chemical Oxygen Demand.

Fig. 4a–c shows the evolution of methane yield, pH and percentage of TCOD reduction respectively for each substrate, during the acclimation period (first 40 days) and after the steady state was reached. In the light of these results, it can be stated that the anaerobic digestion of OMWW without pretreatment is totally unviable at industrial scale, as it inevitably reaches the point of inhibition (i.e., the point at which a rapid decrease in the methane yield, pH and reduction of TCOD is observed) after the first half of the second HRT due to the high inhibitors concentration in the waste. Similarly, OMWW aerated for 7 days also showed inhibition of the anaerobic digestion process after a period amounting to 1.5 times the HRT. Polyphenol concentration in this substrate is lower than OMWW aerated for 5 days, so it is not considered inhibitory, on the other hand the organic load rate is higher, which leads us to believe that the inhibition is due to an organic overload. Thus the anaerobic digestion of OMWW aerated for 7 days for a longer HRT could be possible. By contrast, the anaerobic digestion of OMWW aerated for 5 days was completely stable. The mean values of the main parameters obtained in each experiment after the anaerobic digestion acclimatization period are shown in Table 1.

As for the methane yield, it can be observed that the highest energy generation rate was obtained by the anaerobic digestion of OMWW aerated for 5 days (0.39 m^3 methane/kg TCOD removed), which was 2.4 times higher than that obtained by the anaerobic digestion of untreated OMWW and 1.4 times higher than that obtained by OMWW aerated for 7 days. However, it should be noted that the anaerobic digestion of OMWW

Table 1 – Experimental results anaerobic digestion.

	OMWW without pretreatment	OMWW aerated for 5 days	OMWW aerated for 7 days
m^3 Methane/kg TCOD removed	0.16 ± 0.03	0.39 ± 0.05	0.28 ± 0.03
TCOD substrate (kg/kg TS)	1.20 ± 0.03	0.98 ± 0.01	0.95 ± 0.02
TCOD digested substrate (kg/kg TS)	0.61 ± 0.04	0.35 ± 0.05	0.45 ± 0.03
VFA (g CH_3COOH/L)	8.04 ± 1.38	1.45 ± 1.05	3.24 ± 1.65
Alkalinity (g $CaCO_3$/L)	7.25 ± 1.28	4.64 ± 1.72	6.67 ± 1.82
VFA/alkalinity	1.11	0.31	0.49
pH	6.87 ± 0.13	7.49 ± 0.13	7.20 ± 0.29
VDS	10.92 ± 2.03	7.51 ± 1.62	10.67 ± 1.45
VSS	25.81 ± 2.04	21.17 ± 3.80	35.84 ± 3.34

aerated for 7 days nearly doubled the methane yields obtained by anaerobic treatment of OMWW without pretreatment.

The percentage of TCOD degradation also reached the maximum value through anaerobic digestion of OMWW aerated for 5 days (65.17 ± 5.42%), a considerably high value, given the great polluting power of the initial waste. Aeration pretreatment for 5 days can increase by 31.65% the degradation achieved through anaerobic digestion of non-pretreated OMWW (49.50 ± 3.38%). The level of degradation achieved with the anaerobic treatment of OMWW aerated for 7 days was 52.22 ± 2.73%, a remarkable value, but lower than the level obtained by the biological digestion of OMWW aerated for 5 days.

The best results from an energy and environmental point of view are obtained by treating OMWW aerated for 5 days. This is due to the following reasons:

- The anaerobic digestion process is optimum when the VFA/alkalinity ratio ranges between 0.3 and 0.4, VFA concentration does not exceed 3.00 g/L and pH has a value closer to neutral. These conditions were only found when OMWW aerated for 5 days was anaerobically degraded, making this substrate the only one that did not show inhibition signs during the treatment process.
- The lowest VDS and VSS concentrations, which indicate a higher biodegradation of the waste, were achieved during the anaerobic digestion experiment in which OMWW was aerated for 5 days.

3.3. *Water recovery through anaerobic digestion*

Anaerobic digestion not only produces renewable energy from degradation of highly polluting waste, but also digested effluent (digestate) which, after a decanting process, can generate two high-value products: on the one hand, a sludge that may be subjected to a composting process in order to produce organic amendment, and on the other hand, a liquid effluent that, depending on its physicochemical characteristics, may have several applications (fertirrigation of crops, gardens, public parks, golf courses, pastures, etc.).

Laboratory tests showed that digested effluent from the anaerobic digestion of OMWW aerated for 5 days after a decanting process allowed for the recovery of 20–40% of water and 60–80% of sludge. Therefore, the anaerobic digestion of OMWW aerated for 5 days was observed to recover approximately 30% of the water contained in the waste.

3.4. *Economic viability of the aerobic–anaerobic treatment of OMWW*

The economic viability of an industrial anaerobic digestion plant with a 5-day aerobic pretreatment has been analyzed. It was projected based on the OMWW that would have been generated if the annual amount of TPOMW produced in the province of Badajoz (Spain) had been treated by the company *Troil Vegas Altas S.C.* According to MAGRAMA (2012), 226,700 tonnes of olive were processed in the province by 2009, which generated a total of 56,675 tonnes of TPOWM (Raposo et al., 2004). *Troil Vegas Altas S.C.* generates an amount of OMWW equal to 50% of treated TPOWM (Troil personal communication). The analysis of the biogas plant reported in the present section was scaled to project the treatment of 28,338 annual tonnes OMWW.

The following hypotheses have been made in order to estimate the economic viability of the aerobic–anaerobic treatment of OMWW.

3.5. *Heat requirements of the anaerobic digester*

The anaerobic digester should be maintained at 38 °C. Thus, to satisfy temperature needs, the installation contains a boiler fuelled by the methane generated during the biomethanation. To calculate the heat required to maintain anaerobic digester at 38 °C, the heat necessary to raise the temperature of wastes from 15 to 38 °C was determined and then an additional 62% was added to factor in heat losses in the digester, as well as an additional 5% to ensure adequate operating configuration. Finally, to determine the thermal energy needed to generate the required heat, the calorific value of methane was determined to be 31 MJ/Nm^3, although such value has been somewhere else reported to reach 35 MJ/m^3 (Proyecto Probiogas, 2012), and the efficiency of the biogas boiler was estimated as 80%.

3.5.1. *Installation cost*

Construction costs have been estimated to approach 400,000 € (ESETA France, personal communication). In addition to these, costs regarding feasibility studies (some 10,000 €), administrative needs and authorizations (amounting to 20,000 €) should also be accounted for (Naskeo Environment, 2009).

3.5.2. *Annual benefits*

If the biogas generated in the biological process after postreatment was injected into the natural gas grid, annual benefits will be obtained from the sale of methane at a price of 26.30 €/MWh (CNE, 2011). Only the remaining methane volumes, after heating the substrate to be treated, will be sold.

3.5.3. *Annual cost*

The electricity consumption of the treatment plant was estimated as 7% of the electrical energy that could be generated at a price of 0.12 €/kWh. Additionally, the cost incurred by the operation and maintenance of the anaerobic digestion plant

Table 2 – Economic study of aerobic–anaerobic treatment of OMWW.

OMWW to treat (m^3/year)	28,338
Methane production (Nm^3/year)	899,432
Heat requirement in the anaerobic digestion plant	
Digester demand (GJ/year)	4656
Installation cost	
Anaerobic digestion plant	400,000
Other cost	30,000
Total	430,000
Income	
Injection on methane to the natural gas grid (€)	162,864
Cost	
Consumption of electricity (€)	25,373
Operation and maintenance (€)	8000
Repayment mortgage (€)	43,548
Labour (€)	12,000
Total	88,921
Annual benefits (€)	
73,943	
Period of return on investment (years)	5.82
Net present value (millions of €)	0.29
Internal rate of return (%)	15

was set as 2% of construction costs. Moreover, the annual repayment mortgage was calculated at a projected interest rate of 6% paid over 15 years, and labour costs were assumed as 12,000 €/year.

The results obtained according to these calculations (Table 2) indicate that the aerobic–anaerobic treatment of OMWW is economically feasible given that the Period of Return of Investment (PRI) is 5.82 years, three times less than the lifetime of the installation. In addition, the rest of the economic variables under study showed satisfactory values, namely the Net Present Value (NPV), 0.29 million €, and the Internal Rate of Return (IRR), 15%.

4. Conclusions

(a) Aerobic treatment of OMWW over a period of seven days can achieve the highest decrease in the initial polyphenols concentration (90%).
(b) The optimum aeration time to optimize the biomethanization process is 5 days, since the 5-day aeration substrate was the only substrate that could be treated by anaerobic digestion without showing signs of inhibition.
(c) A reduction in polyphenols concentration of 78% (5 days aeration time), achieves a methane production of 0.39 m^3/kg removed TCOD and 65.17% TCOD degradation.
(d) Anaerobic digestion of OMWW aerated for 5 days can approximately recover 30% of the water contained in the waste, amounting to about 250 L water/m^3 OMWW aerated for 5 days.
(e) The treatment of OMWW by an aerobic–anaerobic digestion is economically profitable.

Acknowledgments

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